Chapter 6. Permutation Tests for Analysis of Variance

Kruskal–Wallis, ordered alternatives, Friedman test, general two-way layouts

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6.1 One-way layout

Motivation. To compare k groups without normality, rank all observations together and compare average ranks across groups; this is the Kruskal–Wallis test, the rank analogue of one-way ANOVA.

Observations $X_{ij} = \mu_i + \epsilon_{ij}$, $i = 1, \dots, k$, $j = 1, \dots, n_i$, $n = \sum n_i$, with i.i.d. errors. We test $H_0 : \mu_1 = \cdots = \mu_k$.

The F-test. With group means $\bar{X}_i$ and grand mean $\bar{X}$,

$$SS_{\text{Tr}} = \sum_i n_i(\bar{X}_i - \bar{X})^2, \qquad SS_E = \sum_{i,j}(X_{ij} - \bar{X}_i)^2, \qquad F = \frac{SS_{\text{Tr}}/(k-1)}{SS_E/(n-k)}.$$

Proposition 6.1

$SS_{\text{Tot}} := \sum_{i,j}(X_{ij} - \bar{X})^2 = SS_{\text{Tr}} + SS_E$. Under H_0 with normal errors, $F \sim F_{k-1,n-k}$.

Proof [Supplement]. Expand $X_{ij} - \bar{X} = (X_{ij} - \bar{X}_i) + (\bar{X}_i - \bar{X})$. The cross term is $2\sum_i(\bar{X}_i - \bar{X})\sum_j(X_{ij} - \bar{X}_i) = 0$. Under normality, $SS_E/\sigma^2 \sim \chi^2_{n-k}$ is independent of the group means. SS_{Tr}/σ^2 is the squared length of the projection of the standard normal vector $(\sqrt{n_i}(\bar{X}_i - \mu)/\sigma)_i$ onto the complement of $(\sqrt{n_i})_i$, hence χ^2_{k-1}. ■

Permutation F-test. Under H_0 (equal distributions) all n observations are exchangeable, so relabel them among the groups: there are $n!/\prod n_i!$ equally likely splits. Since SS_{Tot} is fixed, $F = \frac{n-k}{k-1} \cdot \frac{SS_{\text{Tr}}}{SS_{\text{Tot}} - SS_{\text{Tr}}}$ is increasing in SS_{Tr}, and both give the same test. For moderately large samples the permutation distribution of F is close to $F_{k-1,n-k}$ even without normality.

6.1.1 The Kruskal–Wallis test

Rank all observations jointly. Let $\bar{R}_i$ be the mean rank of group i and S_R^2 the sample variance of all ranks ($= n(n+1)/12$ without ties).

Definition (Kruskal–Wallis statistic)

$KW = \dfrac{1}{S_R^2}\sum_{i=1}^k n_i\Big(\bar{R}_i - \dfrac{n+1}{2}\Big)^2$, which without ties equals $\dfrac{12}{n(n+1)}\sum_i n_i(\bar{R}_i - \frac{n+1}{2})^2$.

Proposition 6.2

(i) The permutation test based on KW coincides with the permutation F-test applied to the ranks.
(ii) Under H_0, $KW \xrightarrow{d} \chi^2_{k-1}$ as all $n_i \to \infty$.

Proof. (i) For ranks, $SS^R_{\text{Tot}} = (n-1)S_R^2$ is fixed and $SS^R_{\text{Tr}} = S_R^2\, KW$, so $F_R = \frac{n-k}{k-1}\frac{KW}{n-1-KW}$ is increasing in KW.

(ii) With $Z_i = \sqrt{n_i}(\bar{R}_i - \bar{R})/S_R$, the without-replacement covariances (Theorem 4.2) give $\text{Cov}(Z) \to I - \mathbf{u}\mathbf{u}'$ with $\mathbf{u} = (\sqrt{n_i/n})_i$, a rank-$(k-1)$ projection. A multivariate finite-population CLT makes Z asymptotically $N(0, I - \mathbf{u}\mathbf{u}')$, so $KW = \|Z\|^2 \xrightarrow{d} \chi^2_{k-1}$. ■

In small samples the χ^2 approximation can be poor in the tails, so the exact or Monte Carlo permutation distribution is preferred.

6.1.2 Ordered alternatives

Against $H_1 : \mu_1 \le \mu_2 \le \cdots \le \mu_k$ (at least one strict), the **Jonckheere–Terpstra** statistic sums the pairwise Mann–Whitney counts:

$$JT = \sum_{i<l} U_{il}, \qquad U_{il} = \#\{(a,b) : X_{ia} < X_{lb}\}.$$

Large JT supports the trend. It equals (up to an affine map) Kendall's τ between the observations and the group index, and its permutation distribution gives an exact test. Using the order information makes it more powerful than KW against monotone trends.

6.2 Randomized complete block design

k treatments in each of b blocks, one observation per cell: $X_{ij} = \mu + t_i + \beta_j + \epsilon_{ij}$. We test $H_0 : t_1 = \cdots = t_k$.

Classical F. $SS_{\text{Tr}} = b\sum_i(\bar{X}_{i\cdot} - \bar{X})^2$ and $SS_E = \sum_{i,j}(X_{ij} - \bar{X}_{i\cdot} - \bar{X}_{\cdot j} + \bar{X})^2$. Under normality, $F = \frac{SS_{\text{Tr}}/(k-1)}{SS_E/((k-1)(b-1))} \sim F_{k-1,(k-1)(b-1)}$.

Why shuffle within blocks. Under H_0, $X_{ij} = \mu + \beta_j + \epsilon_{ij}$: observations in the same block are exchangeable, but observations in different blocks are not (their means differ by block effects). The permutation distribution therefore permutes treatment labels independently within each block, giving $(k!)^b$ equally likely arrangements. Within-block shuffles leave the block means and SS_{Tot} unchanged. Hence $SS_E = SS_{\text{Tot}} - SS_{\text{Blocks}} - SS_{\text{Tr}}$ moves opposite to SS_{Tr}, and the permutation F-test is the same as the permutation test based on SS_{Tr}.

6.2.1 The Friedman test

Rank the k observations within each block. Let $R_i = \sum_j R_{ij}$ be the rank sum of treatment i and $\bar{R}_{i\cdot} = R_i/b$.

Definition (Friedman statistic)

$$FM = \frac{12b}{k(k+1)} \sum_{i=1}^{k} \left(\bar{R}_{i\cdot} - \frac{k+1}{2}\right)^2 = \frac{12}{bk(k+1)} \sum_i R_i^2 - 3b(k+1) \text{ (no ties).}$$

Under H_0 the within-block rank vectors are independent uniform permutations, so the null distribution is distribution-free, and $FM \xrightarrow{d} \chi^2_{k-1}$ as $b \to \infty$. With ties, divide by the average within-block rank variance instead of $k(k+1)/12$.

Example 6.1

$k = 3$, $b = 4$, within-block ranks of (A, B, C): $(1,2,3)$, $(1,3,2)$, $(1,2,3)$, $(2,1,3)$. The rank sums are $5, 8, 11$, so $FM = \frac{12}{48}(25 + 64 + 121) - 48 = 4.5$. The χ^2_2 approximation gives $p = e^{-2.25} = 0.105$. Enumerating all $(3!)^4 = 1296$ arrangements gives the exact $p = 162/1296 = 0.125$. With few blocks the exact distribution is noticeably different.

6.2.2 Page's trend test

For $H_1 : t_1 \le \cdots \le t_k$, use $L = \sum_{i=1}^{k} iR_i$, a sum over blocks of a Spearman-type statistic between treatment order and within-block ranks.

Proposition 6.3

Under H_0, $E(L) = \dfrac{bk(k+1)^2}{4}$ and $\mathrm{Var}(L) = \dfrac{bk^2(k+1)(k^2-1)}{144}$.

Proof. Within one block, $L_j = \sum_i iR_{ij}$ with $(R_{1j}, \ldots, R_{kj})$ a uniform permutation. Each R_{ij} has mean $\frac{k+1}{2}$, so $EL_j = \frac{k+1}{2} \cdot \frac{k(k+1)}{2}$. By Proposition 5.2 applied to $a_i = i$ and $b =$ ranks, $\mathrm{Var}\, L_j = \frac{1}{k-1}\Big(\sum_i (i - \frac{k+1}{2})^2\Big)^2 = \frac{1}{k-1}\Big(\frac{k(k^2-1)}{12}\Big)^2 = \frac{k^2(k+1)(k^2-1)}{144}$. The blocks are independent; multiply by b. ■

6.3 General two-way layouts

With c replicates per cell and no interaction, $X_{ijl} = \mu + t_i + \beta_j + \epsilon_{ijl}$. The permutation test shuffles all kc observations within each block. The rank version (**Skillings–Mack**) ranks within blocks and uses

$$SM = \frac{b\sum_i (\bar{R}_{i\cdot} - \bar{R})^2}{\text{average within-block rank variance}} \dot{\sim} \chi^2_{k-1}.$$

Again, the permutation distribution is more reliable than the χ^2 approximation in small designs. Caution: when interaction is present, a nonlinear transformation such as ranking can destroy or create interaction. Then use special aligned-rank procedures, or treat the model as a regression and use the resampling methods of Chapter 10.